Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub

Model, evidence, and an identification audit

Gabriel Saco

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Universidad del Pacífico

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`github.com/gsaco/ai-03-quispe/tree/branch1`

Roadmap and the paper's contribution

1. Model

Three production modes generate language-specific entry thresholds and an agent-enabled activation band.

2. Evidence

Public GitHub adoption is paired with sharp increases in language breadth, novelty, and entropy.

3. Audit

Test whether the threshold is uniquely agentic and whether adoption timing identifies frontier expansion.

What is new in the paper

It reframes coding AI as a change in the set of feasible production technologies, links that mechanism to language portfolios, and builds a developer-month panel around the public Claude coauthor trailer.

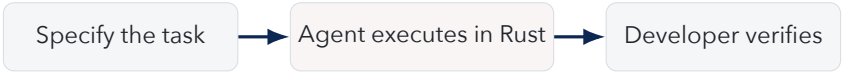
Production frontier is not a skill frontier

Skill frontier

Languages the developer can independently understand, write, debug, and maintain. Expansion here is learning or human-capital accumulation.

Production frontier

Languages in which the developer can ship code using all available tools and collaborators. Expansion here may occur without learning the language.



The outcome is observable production in a language—not demonstrated mastery of that language.

From conversational assistance to delegation

Generation 1: conversation

- human remains the executor;
- suggestions complement language skill;
- interaction is turn-by-turn;
- unfamiliar languages may lack a usable foothold.

Generation 2: agentic delegation

- tool performs multi-step execution;
- human specifies and verifies;
- capability can substitute for language skill;
- residual errors create risk and review costs.

Empirical event

Claude Code's public release creates a setting in which delegated production can be traced through GitHub commit metadata.

Economic environment

Unit of choice

Developer i receives an opportunity in language k at time t with value ω_{ikt}. The language is familiar when skill s_{ik} is high and unfamiliar when it is near a lower bound.

Two kinds of human capital

Language-specific skill supports direct execution. General ability a_i supports task decomposition, specification, and verification across languages.

Technology and costs

Agent capability A_t raises delegated execution. Each mode also carries execution, interaction, compute, verification, and residual-error costs.

Observable portfolio

A language is active when the best available mode yields nonnegative certainty-equivalent surplus. Monthly breadth is the number of active languages.

The developer chooses from a changing menu

Mode choice

$$m^*_{ikt} \in \arg\max_{m \in \mathcal{M}_g} V^m_{ikt}, \qquad Z^g_{ikt} = \mathrm{I}(\max_{m \in \mathcal{M}_g} V^m_{ikt} \geq 0).$$

Before agent adoption

The menu is $\mathcal{M}_1 = \{S, C\}$: solo production or conversational augmentation.

After agent adoption

The menu becomes $\mathcal{M}_2 = \{S, C, D\}$ by adding delegated execution.

Adding a mode cannot shrink the static feasible set; strict expansion requires delegation to dominate at some previously inactive opportunities.

Solo production: skill lowers the entry threshold

Certainty-equivalent surplus

Write solo surplus as opportunity value minus an effective execution cost:

$$V_{ikt}^{S} = \omega_{ikt} - T_{ik}^{S}.$$

Greater language-specific skill lowers T_{ik}^{S} by making implementation faster and safer.

Familiar language

A low solo threshold makes a broad set of opportunities profitable even without AI.

Unfamiliar language

A high solo threshold means only exceptionally valuable projects justify learning and execution costs.

The model treats opportunity value as the source of extensive-margin activation.

Conversational AI: the no-foothold assumption

Augmentation gain

Conversational assistance adds a benefit proportional to language skill and subtracts an interaction cost:

$$V_{ikt}^{C} = V_{ikt}^{S} + \gamma s_{ik} - r_{C}.$$

Assumption 1 for unfamiliar languages

At the lower skill bound, the suggestion gain does not cover interaction cost:

$$\gamma \underline{s} - r_{C} \leq 0.$$

Therefore conversation does not lower the pre-agent entry threshold for an unfamiliar language.

This assumption—not the threshold algebra—is what prevents ordinary chat assistance from producing the same frontier expansion.

Delegation: execution net of verification and risk

Delegated surplus

A compact representation is

$$V_{ikt}^{D} = \omega_{ikt} + \lambda A_t z(a_i) - \kappa(a_i, s_{ik}) - r_D - \rho_{ikt}.$$

Benefits

Agent capability performs execution; general ability improves specification and task decomposition.

Costs

Verification, compute, and residual-error risk can erase the execution gain, especially on difficult or high-stakes tasks.

Delegation expands the frontier only when its net threshold falls below the best pre-agent threshold.

Three thresholds summarize the mode comparison

Pre-agent threshold

Under no foothold, conversation does not beat solo entry for an unfamiliar language, so $T^1 = T^S$.

Delegation threshold

Agent execution matters when $T^D < T^S$ after every verification, compute, and risk cost is included.

Proposition 1: menu expansion and weak frontier growth

Pointwise result

Since the post-agent menu contains every pre-agent option,

$$\max_{m \in \{S,C,D\}} V_{ikt}^{m} \geq \max_{m \in \{S,C\}} V_{ikt}^{m}.$$

Thus an opportunity active before adoption remains feasible afterward in the static model.

From opportunities to language breadth

Summing activation indicators across languages gives weakly greater breadth. Strict growth requires at least one language-opportunity pair for which delegation is profitable while both old modes are not.

The proposition is a menu-expansion theorem; it does not by itself establish why the new mode is available or whether it is used in equilibrium.

Proposition 2: the activation band

Unfamiliar-language result

If conversation has no foothold and delegation strictly lowers the threshold, then

$$Z^2_{ikt} - Z^1_{ikt} = \mathrm{I}(T^D_{ik} \leq \omega_{ikt} < T^S_{ik}).$$

With opportunity distribution F, the activation probability is

$$F(T^S_{ik}) - F(T^D_{ik}).$$

All conditions doing work

No conversational foothold; positive net threshold reduction; opportunity mass in the band; sufficient specification and verification ability; and a stable pool of candidate languages.

Proposition 3: flow, stock, and breadth predictions

New-language flow

A spike at adoption if the agent immediately activates several previously unused languages.

Active breadth

A contemporaneous jump that may persist only while projects continue to use the new languages.

Cumulative stock

A permanent step because every first use remains in the observed historical portfolio.

Interpretive warning

Cumulative languages mechanically retain adoption-month experimentation. Persistent cumulative growth therefore does not prove continued use, mastery, or a permanent productivity effect.

Heterogeneity: specialists and general ability

Specialists

Developers concentrated in few languages have more unfamiliar candidates near the old entry margin. Conditional on comparable opportunities, their first-use response should be larger.

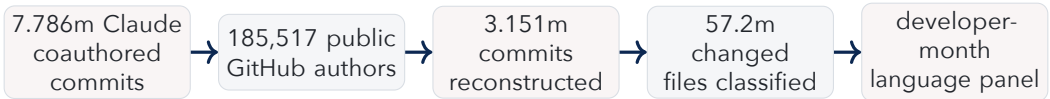
General ability

Higher ability improves specification and verification, reducing delegated cost and residual risk. The paper proxies ability with prior commit volume.

Why the empirical mapping is imperfect

Specialization is measured from observed pre-adoption portfolios and ability from activity. Both can also proxy for project type, available time, employer, or GitHub visibility.

From public trailers to a developer-month panel



Measurement scope

The pipeline observes public commits and file extensions. It does not observe private repositories, non-committed experimentation, prompt content, who wrote each line, or whether the developer understood the code.

Analysis sample and treatment timing

Final panel

- 5,346 developers;
- 2,813 treated early adopters;
- 2,533 later-adopter controls;
- 149,688 developer-month observations;
- January 2024 through April 2026.

Treatment definition

First public commit with Claude's coauthor trailer. Sustained adopters have at least five Claude commits. Screens remove bots and competing coding-agent signals.

Control logic

At each cohort and date, not-yet-treated developers supply the counterfactual comparison.

Four outcome families

Active languages

Distinct programming languages touched by a developer during the month. Captures contemporaneous breadth.

Newly used languages

Languages not previously observed for that developer since January 2024. Captures first-use flow.

Cumulative languages

Total distinct languages ever observed by that month. Captures stock, but mechanically never falls.

Shannon entropy

Dispersion of changed files across languages. Distinguishes balanced diversification from adding one token language.

Repository and commit counts are diagnostic outcomes for the scale of activity, not direct measures of the language frontier.

Empirical strategy: staggered event study

Estimator

Doubly robust Callaway–Sant'Anna group-time effects compare each adoption cohort with developers who have not yet adopted. The specification uses a varying base period and one month of anticipation.

Event window

Six months before through ten months after first public Claude use.

Inference

1,000 bootstrap iterations with clustering at the developer level.

Interpretation

The estimator addresses treatment-effect heterogeneity under staggered timing. It does not make voluntary timing exogenous or absorb a project shock that arrives with adoption.

What the event-study estimand requires

Maintained conditions

- conditional parallel trends;
- no unmodeled anticipation beyond one month;
- stable treatment definition and no interference;
- not-yet adopters remain valid controls;
- outcome histories are measured comparably.

Most demanding condition

In the absence of Claude adoption, early adopters must not experience a coincident project, employer, or workload shock that independently changes language use.

Pre-trends

Flat leads are informative about gradual divergence, but have low power and cannot detect a shock that begins exactly at adoption.

Main estimates: a large adoption-month discontinuity

Outcome	Effect at $e = 0$	Standard error	Scale / interpretation
Active languages	2.528	0.063	pre-adoption mean 0.90
Newly used languages	1.193	0.051	pre-adoption flow 0.31
Language entropy	0.382	0.009	broader file distribution
Cumulative languages	1.604	–	permanent mechanical stock
Commits	35.08	–	strong activity increase
Active repositories	1.494	–	broader project activity

The magnitudes are economically large and precisely estimated; the central question is what produces the adoption-month discontinuity.

Dynamics separate experimentation from persistence

Outcome	Adoption month	One month after	Two months after
Active languages	2.528	1.227	0.693
Newly used languages	1.193	0.126	approximately 0
Cumulative languages	1.604	1.892	2.072
Entropy	0.382	0.189	0.102

What persists

Active breadth and entropy remain positive but decay quickly.

What does not

The flow of first-use languages returns nearly to zero within two months.

Outcome-side robustness checks

Remove first-Claude language

At adoption, active languages still rise by 1.58 and newly used languages by 0.807. The response is not only the language of the first tagged commit.

Remove all Claude commits

In non-Claude-authored commits, active languages rise by 1.663, newly used by 0.724, and cumulative languages by 1.130 at adoption.

Additional checks

Baseline activity controls leave results similar; stricter pre-period activity filters reduce magnitudes but preserve significance; results are not driven by one mechanically tagged commit.

These checks strengthen measurement validity and spillover evidence; they do not independently identify adoption timing.

Mechanism diagnostic: breadth works through volume

Monthly portfolio

Language breadth stays above baseline after adoption because developers make more commits across more repositories and languages.

Per-commit diversity

The number of languages per commit rises mainly in the adoption month and is not persistently higher afterward.

Economic reading

Claude adoption is associated with a scale expansion across the developer's monthly portfolio, not a lasting transformation of each individual commit into a more multilingual object.

Remaining ambiguity

Higher output volume is consistent with lower execution costs, but also with a new deadline, employer, or project that raises both tool demand and coding activity.

Heterogeneity evidence

Specialists respond more

Within activity strata, developers with concentrated pre-adoption portfolios show substantially larger first-language use than generalists. This matches the unfamiliar-language margin.

Illustrative adoption-month effects

High-activity specialists: 0.981 versus 0.301 for generalists. Low-activity specialists: 2.388 versus 1.015.

What it supports

The pattern is consistent with agents relaxing unfamiliar-language constraints rather than merely intensifying an already broad portfolio.

What it does not identify

Specialization and activity are endogenous portfolio characteristics. They need not isolate language skill or general ability from project selection.

Pressure test I: the activation band is generic

Same comparative static without an agent

Let any intervention lower the old entry threshold by $\tau > 0$. Then

$$T^G = T^1 - \tau, \qquad Z^G - Z^1 = \mathrm{I}(T^1 - \tau \leq \omega < T^1).$$

This is exactly an activation band.

Could generate it

Better libraries, an expert collaborator, reusable templates, employer support, or a conventional productivity improvement.

Agent-specific content

The no-foothold and delegation-cost assumptions explain why the new threshold should belong to an agent. The band itself does not.

The model organizes a plausible agent mechanism, but the central threshold theorem is not a uniqueness theorem.

Pressure test II: selection can mimic frontier expansion

Zero-effect counterexample

Let a project shock P_{it} trigger adoption and directly raise language breadth:

$$A_{it} = \mathrm{I}(P_{it} \geq c_i), \qquad Y_{it} = \mu_i + \lambda_t + \beta P_{it} + \tau A_{it} + \varepsilon_{it}.$$

At adoption, the event-study contrast is

$$\tau + \beta\{\mathbb{E}[P \mid \text{adopt}] - \mathbb{E}[P \mid \text{not yet}]\}.$$

It is positive with $\tau = 0$ whenever adopters select on larger project shocks.

Verdict

Mechanism-consistent, not mechanism-identified. The evidence documents a large and robust coincidence between Claude adoption and portfolio expansion, but cannot separate delegation from project selection.

What would be cleaner

Exogenous access or rollout timing; credible instruments; repository-level shocks with unaffected developers; or a comparison against equally large non-agent productivity improvements.

Repository audit

Lean checks the threshold and selection algebra; the simulation produces an adoption jump with a true Claude effect of zero.